

Virtual University of Pakistan

Bt302

Immunology

Final Past Papers

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2018

1. GHVR mechanism ? 5 marks
2. three type of signals need to activate cytotoxic t cells? 5 marks BT302
Final term 11:00am 9 Sep, 2018
1. Three names of vaccines which are live attenuated? 2 marks
2. Name two diseases that are caused by bacterial or viral infection used as vaccine? 2 marks
3. Name types of receptors of cytokines? 3 marks
4. Six Clinical symptoms of GVHR. 5 marks
5. What term is used for type III hypersensitivity? 2 marks
6. Which 2 mechanism involved in Cytotoxic t cells induced apoptosis? 3 marks

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7. Interferon family. 3 marks
8. Hypersensitivity mechanism. 5 marks
9. How a capsular antigen serve as vaccine, which cells are activated by these vaccines, give 1 example. 5 marks
10. Cytokines function, mode of action and receptors types.

2017

1. Anti-body composed of four polypeptide chains **2 identical heavy chains & 2 identical light chains**.
2. After the DNA sequences are cut out and before they are joined, **terminal transferase**, adds some nucleotides to the free ends of the DNAs.
3. B-Cell initially produces IgM (antibody' s heavy chain is encoded by the first constant region gene, the μ **segment**) molecules
4. **Effector cytotoxic T cells** directly kill cells infected with a virus or some other intracellular pathogen.
5. Peptide backbone that bind to invariant pockets formed by conserved amino acids that line **all class II MHC** peptide-binding groove
6. MHC should be able to present peptide fragments from almost any foreign protein to **T cells**
7. The primary risk associated with vaccines, especially vaccines that utilize live organisms, is that the vaccine itself causes **illness**.
8. **Our skin** is a defensive barrier that hinders pathogen invasion. When skin is lost due to injury infection is one of the primary concerns.
9. Hematopoietic stem cell are the stem cells that give rise to all the other **blood cells** through the process of hematopoiesis.
10. **Spleen** Filters out older RBCs
11. Helper T cells stimulate **B cells** to make antibodies that inactivate or eliminate extracellular pathogen & their toxic products.
12. Once a **helper T cell** is activated by an antigen-presenting cell to become an effector cell, it then help activate other cells.
13. The classical pathway is triggered by activation of the **C1-complex (which consists of six molecule C1q and two molecules C1r and C1s)**, either by C1q's binding to antibodies from classes M and G,
14. **Cleavage of C5** by the C3/C5 convertase initiates the assembly of a set of complement proteins that make up the membrane attack complex.
15. the C3b binds to **inhibitory proteins and sialic acid** present on the surface of the body's own cells, and the process is aborted
16. **Skin graft** are rejected faster than other tissue organ kidney or heart
17. **Ag** in the secluded places - are not accessible to the immune system
18. Immune complexes accumulate in many tissues and cause **inflammation and damage or affects many organs or the whole body**
19. IgM or Cold Agglutins- active at **40C**

20. Myasthenia Gravis is caused by auto antibody against **muscle antigen & acetylcholine** receptor antigen.
21. In 1885 the French microbiologist Louis Pasteur and Emile Roux developed the first vaccine against **rabies**.
22. Some vaccines contain killed, but previously **virulent**, micro-organisms that have been destroyed with chemicals, heat, radioactivity or antibiotics.
23. The first such recombinant antigen vaccine approved for human use is the hepatitis **B** vaccine.

24. An **adjuvant** is a pharmacological or immunological agent that modifies the effect of other agents, such as a drug or vaccine.
25. Transplantation is a act of transferring **cells, tissue, or organ** from one site to another
26. **Hyperacute Rejection** which could be induced by prior blood transfusions, multiple pregnancies, prior transplantation, or xenografts.
27. Chronic rejection is characterized by **fibrosis and vascular abnormalities** with loss of graft function over a prolonged period
28. Approximately 30% of all human tumors contain mutated version of **RAS gene**.
29. **T cells** are initially activated in peripheral lymphoid organs by recognizing foreign antigen on the surface of an APC
30. Not true about **Proto Oncogenes**: (See the proto oncogene ppt)
31. RB which overcome the hurdle of **G1 to S hurdle**
32. Risk is that some individuals may have an **allergic reaction** to the vaccine,
33. Grafted tissue has mature **T cells**, they will attack host tissue leading to GVHR
34. **CDKs Phosphorylate crucial target proteins & are expressed constitutively** during cell cycle in inactive form however cyclins are synthesized during specific phases of cell cycle.
35. The binding of CD40 on TH cells toCD40 activates **antigen presenting cells** and induces a variety of downstream effects. 36. **Endocytosis Definition**

Short Question:

- What is proto oncogene (2)
- What type of immune response is elicited by alloantigen (2)
- How MYC produced by MYC mechanism promote cancers (2)
- What is proto oncogene, write 3 types (3)
- Write 6 clinical symptoms of GVHR (3)
- How do proto oncogenes work after activation and write three types of it. (3)

Long Questions:

- Neoplastic transformation and its 4 steps (5)
- How capsular antigen serve as vaccine, which cells are activated by this antigen, give one example (5)
- What are vaccines, write types how does whole organism and exotoxin vaccine induce immune response(10)
- What is auto immunity, write the causes, describe its type in details(10)

Other papers

BT302
what is protogene(2)

what type of immune response is elicited by alloantigen(2)
 how myc produced by myc mechanism promote cancers(2)
 what is proto oncogene, write 3 types(3)
 write 6 symptoms of vhdr(3)
 neoplastic transformation and its 4 steps(5)
 how capsular antigen serve as vaccine, which cells are activated by this antigen, give one example(5)
 what are vaccines, write types how does whole organism and exotoxin vaccine induce immune response(10) what is auto immunity, write the causes, describe its type in details(10)
 BT302 Immunology today's paper:
 Q1: write down two diseases for which inactivated bacterial cells or viral particles are used as vaccines? (2) Marks'
 Q2: What is a proto-oncogene? (2 marks)
 Q3: how does the production of MYC by MYC pathway promote cancer? (2 marks) Q4: how do the proto-oncogenes work after activation? Write down three types of protooncogenes? (3 marks)
 Q5: write down the 6 clinical symptoms of GVHR? (3 marks)
 Q6: write down that how the capsular antigens can serve as vaccines? Which cells can be activated by these antigens? Give one example. (5 marks)
 Q7: how do the acquired environmental conditions bring neoplastic transformation? Write down the four steps. (5 marks)
 Q8: what are vaccines? Write down their type and mechanism by which the whole organism vaccines and inactivated exotoxins (toxoid) induce immunity in the living system? (10 marks)
 Q9: what is autoimmunity? Name the causes of autoimmunity and write down the types of autoimmunity in detail? (10 marks)

Bt302

gene expression/how group1 intron splice dna/afflotxin bi1 lead to replication errors/spliceosome/spliceosome mutation torpedo model crispr structure of protein

Bt302

Cytokines and its types 5 mark Cytotoxic
 t cells 3 marks
 cytokines function, mode of action and receptors types 10 mark
 Three names of disease from attenuated vaccine 3 mark
 Name of oxygen independent antibiotics 3
 What are t cells and two classes of t cells 2 mark

BT302

Define proto oncogene.
 Define hypersensitivity.

Mechanism of hypersensitivity?

What you know about hypersensitivity type3.

Innate and adaptive immunity.

Define antibody and five types of antibody.

Cytokine types .

Vaccines wala tha aik

Bt302

1.Functional Categories of cytokines?

2. signal of cell mediated receptor?

3. Three names of vaccines which are live attenuated?

4. Explain the structure of B cell and T cell under microscope?

5.Mechanism of GVHR?

6. Define antibodies ?and five antibody classes in mammals?

7. Hypersensitivity?

8. Memory cell and Nk cell